

Table 1: DevOps applications in RF and wireless engineering

Application area	Key benefit	Traditional manual approach	Primary tools	DevOps automated approach
RF verification	Immediate feedback on signal quality	Manual bench testing with hardware analysers	GNU Radio, PyTest, NumPy	Automated signal capture and mathematical validation in pipeline
Testbed management	Consistent, reproducible test conditions	Manual cable swapping and hardware config	Ansible, Terraform, Labgrid	Programmable, version-controlled lab environment (IaC)
Field updates	Rapid, low-cost scaling to remote nodes	On-site manual firmware flashing	Yocto, Buildroot, Ansible	Remote OTA (over-the-air) automated deployment
RF simulation	Fail early: catch bad designs before physical build	Ad-hoc simulation and design validation	OpenEMS, Ansys API, Python	'Shift-Left' integration: auto-testing designs before fabrication
Compliance	Reduced risk of regulatory non-compliance	Final certification at end of dev cycle.	Custom test scripts, regulatory libs	'Compliance-as-Code': auto-checking bounds at every commit

The manual work is forwarded to Hardware in the Loop (HIL) for effective and high-performance pipelines. In addition, AI-based platforms are integrated for dynamic intelligence and decision making.

In a DevOps-based pipeline, the following phases are integrated for high performance and minimal error rate:

- **Validation:** To analyse whether the new radio and wireless frequency is within the standards and prescribed limits.
 - **Simulation:** Execution of virtual Software Defined Radio (GNU Radio or similar open source platforms) to analyse the radio and telecom behaviour.
 - **Deployment:** Hardware deployment only if the simulation is working fine and as per the requirements without or with minimal error.
 - **Verification:** Analysis of the signal matching, outputs and confirmation with the target.
- DevOps on AI-based platforms ensures quality and minimal errors in infrastructure deployment. In

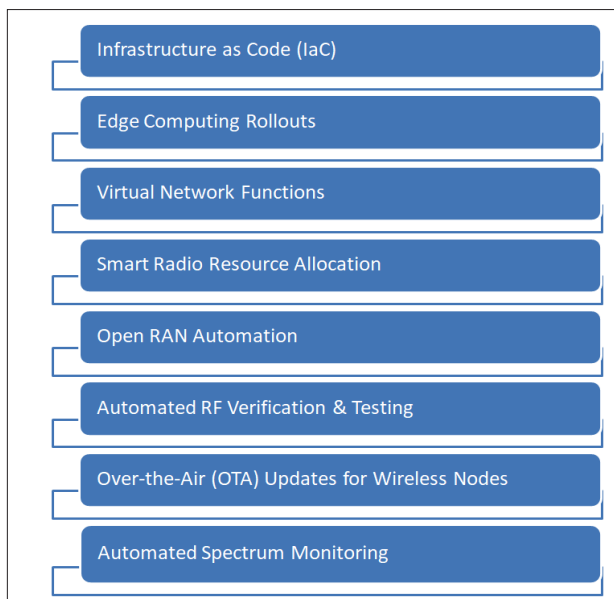


Figure 1: Key use cases of DevOps in radio frequency, telecom and emerging wireless technologies

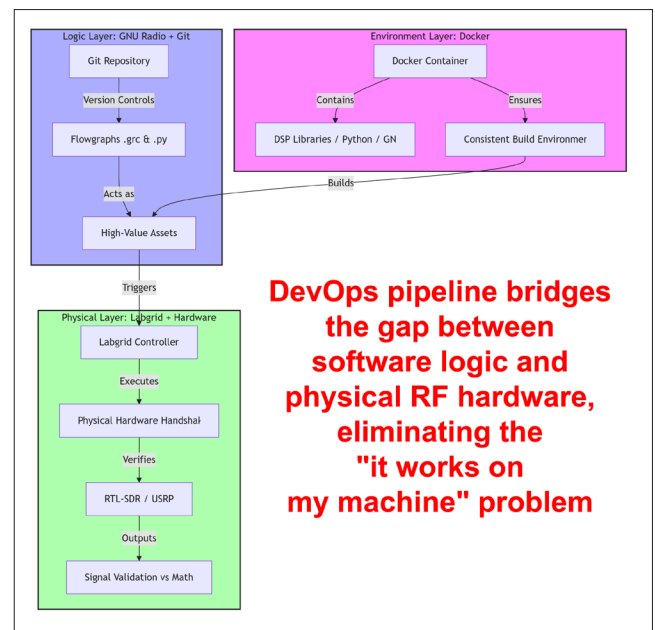


Figure 2: DevOps pipeline for software logic and RF hardware